# Kartik Makkar

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## Summary

**AI Research Engineer & Software Architect** with 14+ years building enterprise platforms and production AI systems. Currently cost & efficiency lead for Cisco's **Circuit/MyAgent** enterprise AI platform — model routing and cascading, OSS-model adoption, and the agent-evaluation infrastructure behind model selection. Previously architected AssessmentX AI, a healthcare AI platform deployed across 40+ hospital systems and 400+ assessment projects. Combines agent evaluation, hybrid retrieval, multi-agent orchestration, MCP, RAFT/LoRA fine-tuning, and AI observability with production architecture across Python/FastAPI, Postgres, Java/Spring, and cloud-native systems. **M.S. Computer Science (AI), Georgia Tech, 2024.**

## Independent Research & Open-Source Systems

**MAKBench** — agent benchmarking & cost-observability harness with a published leaderboard  Code  Leaderboard

- Hermetic per-attempt Docker sandboxes with an LLM gateway as the single source of cost truth (budget-capped virtual keys — metered, never self-reported), split-phase grading by a trusted verifier the agent never sees, and anti-triviality checks; live leaderboard covering **2 agents** × **7 models** × **10 enterprise tasks** with pass@1, **pass^k** reliability, and cost-of-success.

**HCFT** — healthcare-domain 12B RAFT fine-tune for grounded RAG and calibrated refusal  Code  Write-up

- Trained a full-precision **bf16 LoRA** on 8×H200 GPUs (DDP) with RAFT data generated from a frozen hybrid retriever (dense + BM25 + RRF + rerank; hit@5 ∼0.78); under a bias-controlled three-model-family judge panel the 12B **tied GPT-5.4-mini head-to-head** and led calibration at **79.5% balanced accuracy vs. 67.4% / 57.3%**, refusing **74% of unanswerable questions vs. 16%** for the full frontier model.

**Laika** — BYOK coding agent for VS Code *(in active development)*  Code

- Provider-agnostic coding agent (host-agnostic core runtime, VS Code extension host, React sidebar) with **tiered autonomy** (manual / guarded / autonomous) governed by a per-repo policy file, and "Orbit" — a live overview of the in-flight change set.

**Conclave** — multi-expert LLM deliberation rooms  Code

- Multi-expert deliberation where **every expert drafts its own solution from scratch, then proposes and critiques changes with rationale — the way developers review a PR — and the merged edits execute only once the room converges**; FastAPI + async SQLAlchemy with Postgres as system of record, job queue (`FOR UPDATE SKIP LOCKED`), pause signal, and event feed.

## Experience

**Cisco Systems** — **Nov 2018 – Present**
*San Francisco Bay Area, CA*

**AI Research Engineer | Cost & Efficiency Lead, Circuit/MyAgent Platform** — **Jul 2026 – Present**

*Technical leadership on Cisco's enterprise AI team, owning cost, efficiency, and model optimization — OSS and lower-cost model adoption via routing and cascading — across the **Circuit/MyAgent** platform (enterprise Q&A + the **Cisco MyAgent** agent).*

- **Agent evaluation program and platform. Qualified a faster, lower-cost alternate model and secured MyAgent's on-schedule launch** through a model-agnostic evaluation spanning 1,400+ workflows and live MCP tool use — **70,651 trials (828,206 agent steps)** across frontier and in-house models with fault injection, LLM-as-judge scoring, and per-trial quality/latency/cost telemetry — productionized as a team-scale platform (FastAPI, Postgres-backed job queue, React live monitoring, 51-chart Superset analytics).
- **Continuous model qualification.** Leading controlled control-vs-experiment studies — a baseline agent paired against a Rust-based experimental target — that qualify candidate models across short- and long-running task workflows and extended conversations under identical prompts, tools, and scoring; the evidence base for the platform's routing, cascading, and deployment decisions.

**Software Architect – AssessmentX AI Platform** — **Nov 2024 – Jun 2026**

*Hands-on architect for AssessmentX AI, Cisco's production platform for knowledge ingestion, structured and semantic search, grounded chat, and multi-skill agentic analysis across 40+ hospital systems and 400+ assessment projects. Defined shared architecture across agent orchestration, model routing, retrieval, evaluation, observability, governance, and developer tooling.*

- **Reusable agent platform.** Built a domain-agnostic LangGraph SDK powering three production agents through declarative skill manifests, five graph templates, three-tier model resolution, Pydantic-validated outputs, and bounded retries — replacing bespoke plan/act/validate implementations.
- **Multi-intent retrieval and data agent.** Designed a schema-aware planner that routes each turn across grounded RAG, NL-to-MongoDB aggregation, or multi-step analysis. Combined vector and BM25 retrieval with RRF and BGE/LLM reranking; reduced the route/classify/plan chain from ∼3 LLM calls to one, lowering per-turn token cost.
- **Supervisor analysis runtime.** Architected parallel specialist fan-out with structured outputs, bounded recovery, and live SSE topology traces; reduced manual research by **40–50 hours per customer engagement**.
- **Governed spec-to-code agent.** Built markdown-to-Python assessment generation with AST allowlisting, SHA-256 human approval, immutable snapshots, and memory/time-capped sandbox execution; cut delivery time for a new assessment type from **∼4 weeks to ∼25 minutes** and reduced code-generation cost through model routing.
- **AI observability and production safety.** Created self-hosted HIPAA-aware DAG/span tracing, token and cost attribution, LLM-as-judge scoring, and run comparison; preserved data sovereignty, removed **∼$20K/year in cross-charges**, and enforced grounding, refusal, and tenant isolation.

- **Agentic SDLC framework.** Established ∼55 reusable multi-IDE skills, an age-encrypted secret vault, a TypeScript MCP server (31 tools / 10 resources / 5 prompts), and CSDL/OWASP security gates for AI-native delivery.

**Software Engineer IV – Healthcare Assessment Platform** **Sep 2022 – Nov 2024**

*Built a healthcare assessment-automation platform that compressed weeks of solution-architect effort into minutes of repeatable ingestion, scoring, recommendations, and deliverable generation for hospitals, providers, and life-sciences customers.*

- **Assessment automation platform.** Architected and shipped the platform end-to-end across Java 21 / Spring Boot 3.5 microservices, a Python AI layer, and Angular 18; automated specification ingestion, third-party HIMSS scoring intake, score computation, dashboards, and DOCX/PPTX deliverables.
- **Priority-aware recommendations.** Led parallel workstreams across ingestion, data-collection UI, scoring, dashboards, deliverables, and practitioner workflows; modeled personas, themes, services, use cases, and leadership priorities to align recommendations with business goals.
- **Cross-stack integration and security.** Unified Java and Python services and project lifecycle workflows, ported Flask to FastAPI, moved secrets to AWS Secrets Manager, and drove CSDL, SonarQube, OAuth, CSP, and OWASP ZAP remediation.

**Software Engineer IV – AppDynamics Customer Engineering** **Feb 2022 – Sep 2022**

- Architecture-level observability SME embedded with enterprise customers: mapped application architecture, defined KPIs with engineering leaders, and supported delivery including custom AppDynamics plugins.

**Software Engineer III/IV – Collaboration Platform Engineering** **Nov 2018 – Feb 2022**

- Engineering lead on Cisco's Collaboration Workflow Manager: owned data-extraction and aggregation microservices, drove cluster consolidation, data migration, and rollouts for a Big Four U.S. bank and a global technology company; shipped a full-stack Webex Contact Center analytics dashboard (Angular, Spring Cloud, MongoDB).

**Infosys** **Aug 2011 – Nov 2018**

*Bangalore, India & San Francisco Bay Area*

**Full-Stack Engineer** **Aug 2011 – Nov 2018**

- Engineered the workflow-automation core of an enterprise provisioning platform for a global bank (jBPM directed-graph orchestration of Java services, Spring REST/SOAP APIs; 500K+ requests over five years); later technical lead driving the monolith-to-microservices re-architecture for a U.S. technology client.

## Education

**M.S. Computer Science (Artificial Intelligence)**, *Georgia Institute of Technology* — Atlanta, GA **Dec 2024**

**B.Tech. Electronics & Communications**, *Punjabi University, Patiala* — Punjab, India **Jan 2011**

## Technical Skills

**AI / ML — Systems & Retrieval:** Agent SDK development, multi-agent orchestration, LangGraph state machines, MCP, ReAct, MCP tool-use benchmarking, LLM codegen with AST gating, hybrid search (dense + BM25 / TF-IDF), RRF, cross-encoder / LLM reranking, hierarchical retrieval, NL-to-MongoDB / NL-to-SQL, GraphRAG, embeddings

**AI / ML — Modeling, Evaluation & Safety:** Fine-tuning SLMs / LLMs, LoRA / QLoRA, RAFT, synthetic data generation, RAG evaluation, LLM-as-judge, pairwise evaluation, RAGAS, agent harness design, calibration and refusal analysis, fault injection, AI guardrails, grounding / hallucination control, prompt-injection defense, tenant isolation

**AI / ML Tools:** PyTorch, Hugging Face Transformers, TRL, PEFT, LangChain, LangSmith, Databricks, MLflow, Modal, pandas, NumPy, scikit-learn, XGBoost, TensorFlow, AWS Bedrock, AWS SageMaker, GPT-4o, Claude, Gemini, Llama, Azure OpenAI

**Backend & Databases:** Python (FastAPI / Flask), Java 21, Spring Boot 3 / Spring Cloud, Maven, REST / gRPC / SSE, microservices, distributed systems, event-driven architecture, system design, MongoDB Atlas, PostgreSQL, DuckDB, Apache Superset

**Frontend:** React, Angular 18, TypeScript, standalone components, signals

**Security:** HIPAA-aware design, CSDL compliance, OWASP ZAP, Trivy, Semgrep, Anchore, OAuth, JWT, AWS Secrets Manager, age-encrypted secret vault

**Development & Cloud:** Cursor, Claude Code, GitHub Copilot, Windsurf, Conda, Docker, Kubernetes, OpenShift, Terraform, CI/CD, Jenkins, AWS (S3 / ECR / Secrets Manager / OpenSearch / EC2 / IAM), GCP, Azure, Jira, Confluence, SharePoint, Splunk